

Normalization (1NF to 3NF): Sample Problems with Solutions

Database Systems Assignment

Problem 1: Library Book Holdings

Scenario: A library tracks books and their authors. One book can have multiple authors.

BookID	Title	Authors
B1	Database Systems	Elmasri, Navathe
B2	Operating Systems	Silberschatz

Tasks:

- (a) Is this relation in 1NF? Justify your answer.
- (b) Convert the relation to 1NF.

Solution:

- No, this table is not in 1NF because the **Authors** attribute contains multiple values (not atomic).
- To convert to 1NF, separate each author into its own row:

BookID	Title	Author
B1	Database Systems	Elmasri
B1	Database Systems	Navathe
B2	Operating Systems	Silberschatz

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Problem 2: Course Registration

Scenario: A university tracks course registration using:

StudentID	CourseID	StudentName	CourseName
1001	CS101	Alice	Algorithms
1002	MA102	Bob	Calculus

Functional Dependencies:

- $\text{StudentID} \rightarrow \text{StudentName}$
- $\text{CourseID} \rightarrow \text{CourseName}$
- $(\text{StudentID}, \text{CourseID})$ is the composite primary key

Tasks:

- Is this table in 2NF? Explain.
- Normalize it to 2NF.

Solution:

- No, it is not in 2NF due to partial dependencies:
 - StudentName depends only on StudentID.
 - CourseName depends only on CourseID.
- Decomposition into 2NF:
 - **Students(StudentID, StudentName)**
 - **Courses(CourseID, CourseName)**
 - **Enrollments(StudentID, CourseID)**

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Problem 3: Movie Review System

Scenario: This relation stores user movie reviews:

ReviewID	UserID	UserName	MovieID	MovieTitle
R1	U1	Alice	M1	Inception
R2	U2	Bob	M2	Interstellar

Functional Dependencies:

- $\text{ReviewID} \rightarrow \text{UserID}, \text{MovieID}$
- $\text{UserID} \rightarrow \text{UserName}$
- $\text{MovieID} \rightarrow \text{MovieTitle}$

Solution:

- **Transitive dependencies:**
 - $\text{ReviewID} \rightarrow \text{UserID} \rightarrow \text{UserName}$
 - $\text{ReviewID} \rightarrow \text{MovieID} \rightarrow \text{MovieTitle}$
- Decompose into 3NF:
 - **Reviews(ReviewID, UserID, MovieID)**
 - **Users(UserID, UserName)**
 - **Movies(MovieID, MovieTitle)**

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Problem 4: Order Management System

Scenario:

OrderID	CustomerID	CustomerName	ProductID	ProductName	Quantity
O1	C1	Alice	P1	Pen	3
O2	C2	Bob	P2	Notebook	2

Functional Dependencies:

- $\text{OrderID} \rightarrow \text{CustomerID}, \text{ProductID}, \text{Quantity}$
- $\text{CustomerID} \rightarrow \text{CustomerName}$
- $\text{ProductID} \rightarrow \text{ProductName}$

Solution:

- Violates 3NF because:
 - $\text{OrderID} \rightarrow \text{CustomerID} \rightarrow \text{CustomerName}$ (transitive)
 - $\text{OrderID} \rightarrow \text{ProductID} \rightarrow \text{ProductName}$ (transitive)
 - Decomposition:
 - **Orders(OrderID, CustomerID, ProductID, Quantity)**
 - **Customers(CustomerID, CustomerName)**
 - **Products(ProductID, ProductName)**
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Problem 5: Library Book Loans

Relation: Loan(BookID, MemberID, MemberName, BookTitle, LoanDate)

Functional Dependencies:

- $(\text{BookID}, \text{MemberID}) \rightarrow \text{LoanDate}$
- $\text{MemberID} \rightarrow \text{MemberName}$
- $\text{BookID} \rightarrow \text{BookTitle}$

Solution:

- Violates 2NF and 3NF due to partial and transitive dependencies.
- Decompose into 3NF:
 - **Loan(BookID, MemberID, LoanDate)**
 - **Members(MemberID, MemberName)**
 - **Books(BookID, BookTitle)**